

Dohwan Ko

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EDUCATION	Korea University M.S. & Ph.D. integrated in Computer Science and Engineering Advisor: Hyunwoo J. Kim	Seoul, Republic of Korea Mar. 2021 – Present
	Korea University B.S. in Computer Science and Engineering	Seoul, Republic of Korea Mar. 2017 – Feb. 2021
RESEARCH INTERESTS	• Multi-Modal Agent AI and Reasoning • Video Large Language Models and Foundation Models • Efficient Multi-Modal Video Understanding	
WORKING EXPERIENCE	Meta <i>Research Scientist Intern, Meta Reality Labs</i>	Menlo Park, CA April 2026 – Present
	• Enhancing a vision-language model with adaptive latent reasoning using model looping and internal thinking tokens. • Collaborators: Patrick Huber, Ernie Chang, Adithya Sagar, Chinnadhurai Sankar, Pierce Chuang	
	Meta <i>Research Scientist Intern, Meta Reality Labs</i>	Menlo Park, CA Mar. 2025 – Jun. 2025
	• Developing a streaming video memory agent for query-agnostic online frame selection in online video understanding. • Collaborators: Pierce Chuang, Debojeet Chatterjee, Ashish Shenoy, Yichao Lu, Shane Moon, Luna Dong, Vikas Bhardwaj.	
	NEC Labs America <i>Research Scientist Intern, Media Analytics Team</i>	San Jose, CA Jun. 2024 – Aug. 2024
	• Developing a vision-language model enhanced with kinematic instruction tuning for spatio-temporal reasoning over object speed, movement direction, and traveled distance. • Collaborators: Yumin Suh, Vijay Kumar B G, Manmohan Chandraker.	
	NAVER <i>Engineering Intern, User Feedback Platform</i>	Seongnam, Republic of Korea Jul. 2020 – Aug. 2020
	• Building an AI model for detecting malicious comments.	
PUBLICATIONS	[1] SVMemAgent: A Streaming Video Memory Agent for Query-Agnostic Online Frame Selection <u>Dohwan Ko</u>, Ji Soo Lee, Pierce Chuang, Debojeet Chatterjee, Ashish Shenoy, Yichao Lu, Shane Moon, Luna Dong, Vikas Bhardwaj, Hyunwoo J. Kim <i>under review.</i> [2] VIDEOSEARCH-R1: Iterative Video Retrieval and Reasoning via Soft Query Refinement Seohyun Lee*, Seoung Choi*, <u>Dohwan Ko</u>*, Jongha Kim, Hyunwoo J. Kim European Conference on Computer Vision (ECCV), 2026. [paper] [3] MoE-GRPO: Optimizing Mixture-of-Experts via Reinforcement Learning in Vision-Language Models	

Dohwan Ko, Jinyoung Park, Seoung Choi, Sanghyeok Lee, Seohyun Lee, Hyunwoo J. Kim
Computer Vision and Pattern Recognition (CVPR), 2026. [paper]

- [4] **DocPrune: Efficient Document Question Answering via Background, Question, and Comprehension-aware Token Pruning**
Joonmyung Choi, Sanghyeok Lee, Jongha Kim, Sehyung Kim, Dohwan Ko, Jihyung Kil, Hyunwoo J. Kim
Computer Vision and Pattern Recognition (CVPR), 2026. [paper]
- [5] **Bidirectional Likelihood Estimation with Multi-Modal Large Language Models for Text-Video Retrieval**
Dohwan Ko*, Ji Soo Lee*, Minhyuk Choi, Zihang Meng, Hyunwoo J. Kim.
International Conference on Computer Vision (ICCV) **Highlight** (top 9.7%), 2025. [paper]
- [6] **ST-VLM: Kinematic Instruction Tuning for Spatio-Temporal Reasoning in Dynamic Videos**
Dohwan Ko*, Sihyeon Kim*, Yumin Suh, Vijay Kumar B G, Minseo Yoon, Manmohan Chandraker, Hyunwoo J. Kim
Preprint, 2025. [paper]
- [7] **LLaMo: Large Language Model-based Molecular Graph Assistant**
Jinyoung Park, Minseong Bae, Dohwan Ko, Hyunwoo J. Kim.
Advances in Neural Information Processing Systems (NeurIPS), 2024. [paper]
- [8] **Large Language Models are Temporal and Causal Reasoners for Video Question Answering**
Dohwan Ko*, Ji Soo Lee*, Wooyoung Kang, Byungseok Roh, Hyunwoo J. Kim.
Empirical Methods in Natural Language Processing (EMNLP), 2023. [paper]
- [9] **Open-Vocabulary Video Question Answering: A New Benchmark for Evaluating the Generalizability of Video Question Answering Models**
Dohwan Ko, Ji Soo Lee, Miso Choi, Jaewon Chu, Jihwan Park, Hyunwoo J. Kim.
International Conference on Computer Vision (ICCV), 2023. [paper]
- [10] **MELTR: Meta Loss Transformer for Learning to Fine-tune Video Foundation Models**
Dohwan Ko*, Joonmyung Choi*, Hyeong Kyu Choi, Kyoung-Woon On, Byungseok Roh, Hyunwoo J. Kim.
Computer Vision and Pattern Recognition (CVPR), 2023. [paper]
- [11] **Randomly Shuffled Convolution for Self-Supervised Representation Learning**
Youngjin Oh*, Minkyu Jeon*, Dohwan Ko, Hyunwoo J. Kim.
Information Sciences, 2022. [paper]
- [12] **Video-Text Representation Learning via Differentiable Weak Temporal Alignment**
Dohwan Ko, Joonmyung Choi, Juyeon Ko, Shinyeong Noh, Kyoung-Woon On, Eun-Sol Kim, Hyunwoo J. Kim.
Computer Vision and Pattern Recognition (CVPR), 2022. [paper]
- [13] **Search-and-Attack: Temporally Sparse Adversarial Perturbations on Videos**
Hwan Heo*, Dohwan Ko*, Jaewon Lee*, Youngjoon Hong, Hyunwoo J. Kim.
IEEE ACCESS, 2021. [paper]

TALKS AND
SEMINARS

Transformer and Visual Question Answering
Korea University

Nov. 2023

TEACHING
EXPERIENCE

Deep Learning (COSE474)
Teaching Assistant at Korea University
Artificial Intelligence (COSE361)
Teaching Assistant at Korea University

Seoul, Republic of Korea
Sep. 2022 - Dec. 2022

Seoul, Republic of Korea
Mar. 2022 - Jun. 2022

HONORS &
AWARDS

Outstanding Paper Award, Korea University

2024

	Semester High Honors , Korea University	2020
	Veritas Program Scholarship , Korea University	2017
ACADEMIC SERVICES	Reviewer of ICLR (International Conference on Learning Representations)	2026
	Reviewer of NeurIPS (Advances in Neural Information Processing Systems)	2025-2026
	Reviewer of ICCV (International Conference on Computer Vision)	2025
	Reviewer of CVPR (Computer Vision and Pattern Recognition)	2024-2026
	Reviewer of ECCV (European Conference on Computer Vision)	2024
	Reviewer of ACL ARR (Association for Computational Linguistics Rolling Review)	2024-2025
SKILLS	Language	
	• Korean (<i>native</i>)	
	• English (<i>intermediate</i>)	
	Programming C, Python, PyTorch, git, $\LaTeX$	